AMRITA VISHWA VIDYAPEETHAM
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SCHOOL OF
ARTIFICIAL INTELLIGENCE
COIMBATORE

Project Title

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Team Number: AB03

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Introduction & Motivation

- Public procurement is a vital process that enables governments to procure goods and services through transparent and competitive bidding.
- Micro, Small, and Medium Enterprises (MSMEs) play a significant role in economic growth but often face challenges in participating successfully in government tenders.
- Existing e-procurement platforms primarily rely on rule-based workflows and offer limited intelligent support for tender analysis, compliance verification, and fraud detection.
- Recent advancements in Agentic AI, Large Language Models (LLMs), Machine Learning, and Graph Analytics enable the development of intelligent systems capable of automating complex procurement tasks.
- This project proposes an Agentic AI-powered Procurement Intelligence Platform that leverages multiple specialized AI agents to assist MSMEs, improve procurement efficiency, and strengthen transparency and security in public procurement.

Literature Review

PAPER	METHODOLOGY	RESULT	LIMITATION
A Multi-Agent Large Language Model Framework for Intelligent Vendor Evaluation and Risk aware Procurement decision.(2026)	Multi-Agent LLM	Better procurement decisions	No behavioral security
Fraud, Corruption and Collusion in public procurement in Public procurement Activities . (2022)	Graph Analytics	Detects cartels	No compliance assistence

Research Gap

- 2–3 precise gaps identified from literature.

Problem Statement

Current e-procurement systems provide limited intelligent assistance for tender preparation and rely primarily on rule-based validation. As a result, many MSMEs face technical bid rejections due to complex tender documents, compliance errors, and manual verification, while procurement authorities lack effective AI-driven tools to assess financial risk, detect fraudulent bidding activities, and support transparent decision-making.

Objectives

Primary Objective

- Develop an Agentic Multi-Agent Procurement Platform for intelligent procurement assistance, compliance verification, and risk assessment.

Secondary Objectives

- Automate Tender Compliance: Extract tender requirements using LLMs and validate them against MSME profiles.

- AI-Based Financial & Risk Analysis: Evaluate bidder financial health and predict procurement risks using OCR and ML.

- Intelligent Fraud Detection: Identify cartels, suspicious bidding patterns, and bot activities using Graph AI and Behavioral AI.

- Explainable Procurement Intelligence: Analyze tender amendments and provide transparent AI-driven recommendations.

Proposed Methodology / Framework

- Insert block diagram or flowchart of the full pipeline.

Techniques & Tools

- Algorithms, datasets, software, hardware, platforms.

Expected Outcomes

- Prototype/model/result and performance metrics.

Industry Relevance[Optional]

- Use-case, sector alignment, partners/MoUs if applicable.

Timeline

- Semester-wise plan, milestones, deliverables.

Team Roles

- Member-wise responsibilities and modules.

References

- IEEE-format references matching citations.